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The power input ratings of industrial applications for the compressors vary between 2000 and 90,000 kW. (See Fig. C-142.)

With a view to economical manufacturing and stocking of the major components, the range of compressors has been standardized. Most OEMs stock major components. That facilitates prompt delivery of machine parts such as rotor blades and stator blades, bearings, joints, etc., for service requirements. The systematic design of components over the whole range of sizes enables the compressor to be adapted optimally to the required operating conditions. Measurements conducted on the blading of various designs and sizes ensure exact conformity of the design data with the operating conditions.

Operating range. OEMs compete to offer the widest range of operating parameters. How they do so is specific to their individual design philosophies. This information source's axial compressor program comprises two type ranges:

Type A. Compressors with fixed stator blades (FIXAX) (see Fig. C-131).

Type AV. Compressors with adjustable stator blades on all or only some stages at the inlet (VARAX) (see Fig. C-131).

Each type range consists of 12 geometrically graduated sizes with rotor diameters extending from 40 to 140 cm. This completely covers a suction volume range of 70,000 to 1,250,000 m^3/h. (See Fig. C-143.)

The required compressor size and number of stages, together with the corresponding standardized overall length, are selected according to the suction volume and the thermodynamic head.

Type A (FIXAX). The A-type FIXAX compressor is generally used whenever the driver is a steam turbine, a split-shaft gas turbine, or a variable-frequency high-speed synchronous motor.

The required operating points can be attained by speed variation, and there is basically no need for adjustable stator blades. Fixed-blade machines are also selected for installations where only minor flow variations are required, or if the mass flow is adapted by variation of the suction pressure, as in aerodynamic test facilities, for example.

Type AV (VARAX). The AV series with adjustable stator blades permits a large stable operating range at constant speed. It is therefore used for constant-speed electric motor drive. Nevertheless, this type is being increasingly preferred for steam turbines and single-shaft gas turbines as well. In this particular case, the stator blade control either facilitates operation with limited speed control range (increased reliability of operation for certain turbine types) or, in combination with the speed control, provides an additional extension of the operating range and an improvement of the overall efficiency at part load. Furthermore, it offers the advantage of quick adaptation of the compressor to changed operating conditions without acceleration of the set—a characteristic that is of great interest for the periodic charging of air heaters in blast-furnace blowing plants. Many axial compressors are fitted with adjustable stator blades.

Stator blade setting with electric servomotor. For a great number of processes, the reference value of pressure or mass flow is selected at the process control panel and transmitted to the compressor servomotor. In this case this servomotor is of the electric type with the additional possibility of manual adjustment of the stator blades.

Stator blade control with hydraulic servomotor. If the process calls for automatic pressure or mass flow control, the stator blade adjusting mechanism will be operated by a synchronized pair of hydraulic servomotors.

With the exception of the stator blades and their adjusting mechanism, the same standardized construction elements are used for both FIXAX and VARAX types.

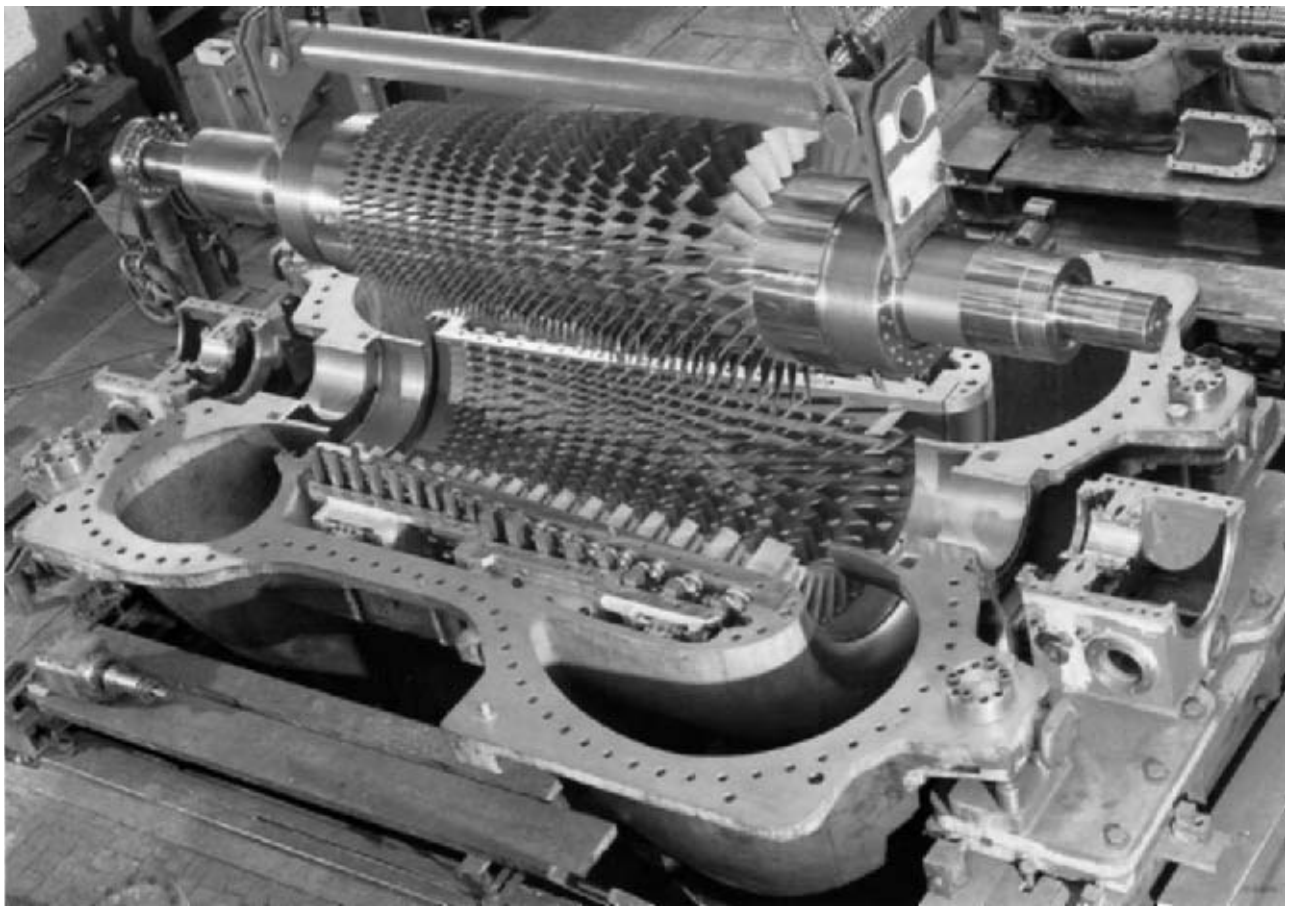
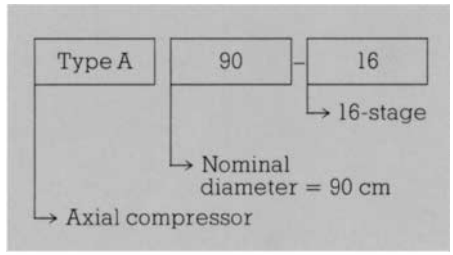


FIG. C-142 Axial compressor, type AV 100–16, during erection. Two identical steam-turbine–driven machines are supplying air to the blast furnace of a British steel works. Suction volume 560,000 Nm^3/h, discharge pressure 6.2 bar, power input 52,000 kW each. (Source: Sulzer-Burckhardt.)

Performance data. Figures C-144 through C-149 (note type designation) facilitate the selection of the

Compressor size	Nominal diameter	D (cm)
	Number of stages	z (–)
	Power input	P (kW)
	Speed	n (rev/min)
	Discharge temperature	T_2 (K)/t_2 (°C)
Using	Capacity	$\dot{m}$ (kg/s)
	Suction pressure	p_1 (bar abs)
	Suction temperature	T_1 (K)/t_1 (°C)
	Relative humidity of the air or gas	φ_1 (%)
	Discharge pressure	p_2 (bar abs)
	Molecular mass	M (kg/kmol)
	Isentropic exponent	k (–)
	Compressibility factor	Z (–)

The following factors and symbols are also used for the calculation:



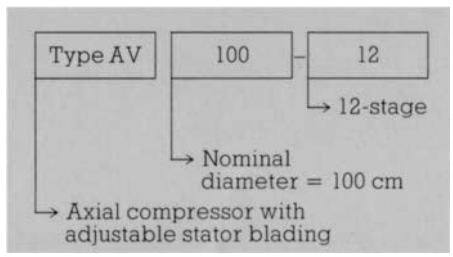


FIG. C-143 Type designation. (Source: Sulzer-Burckhardt.)

	Suction volume (actual)	$\dot{V}_1$ (m³/s)
	Absolute humidity	x (–)
	Polytropic efficiency	η_P (%)
Indices	Suction branch	1
	Discharge branch	2
	Dry	t
	Wet	f

Design features. The basic design offers (see Fig. C-150):

- Cast casing and separate blade carrier
- Casing supported by means of pendulum supports (minimum expansion forces) or feet with keyways for the smaller frame sizes
- Solid or hollow rotors, smooth-running characteristics with integrated balancing pistons
- Blading with optimal aerodynamic characteristics—with good efficiency and specific capacity, low stressing, favorable control characteristics
- Blade vibrations minimized; blade profiles and blade fixation are designed accordingly
- Adjustable stator blades—of standard design—for optimum flow control
- Maintenance-free, oil-free stator blade adjusting mechanism that is also protected against the ingress of contaminants
- Various bearing options
- Various shaft seal options

Casings, bearing pedestals. Depending on the type of gas and the design pressure, the casings are made of gray cast iron, nodular cast iron, or cast steel. The cast design incorporates a rigid construction, effective noise attenuation, and aerodynamically favorable layout of the respective ducting. The suction and delivery branches are usually routed vertically downward. In cases where, due to the composition of the gas and/or the pressure level, steel casings are mandatory, a welded construction

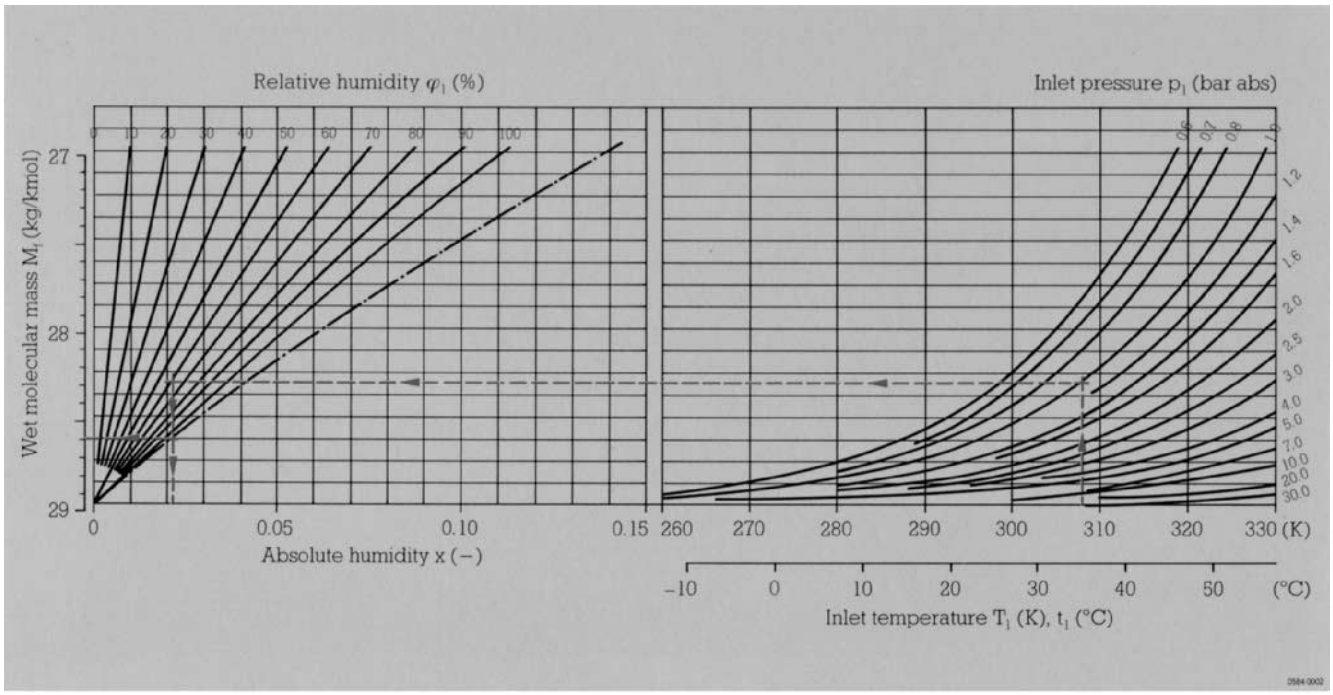


FIG. C-144 Determination of the absolute humidity *x* and the molecular mass M_f of the wet gas. (Source: Sulzer-Burckhardt.)

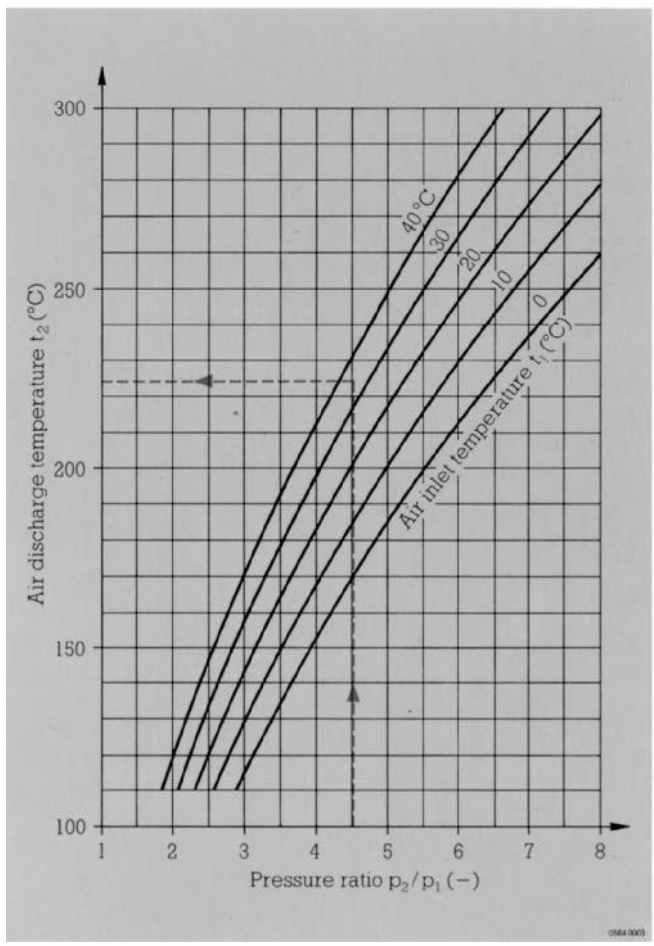
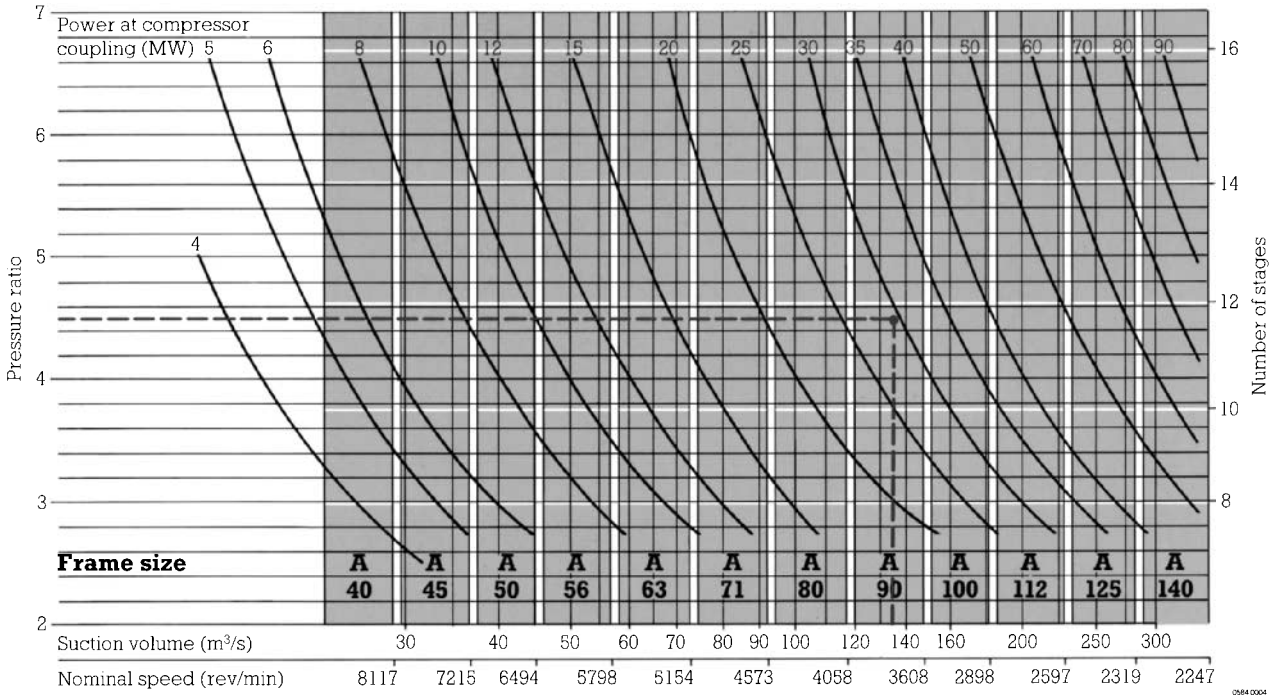


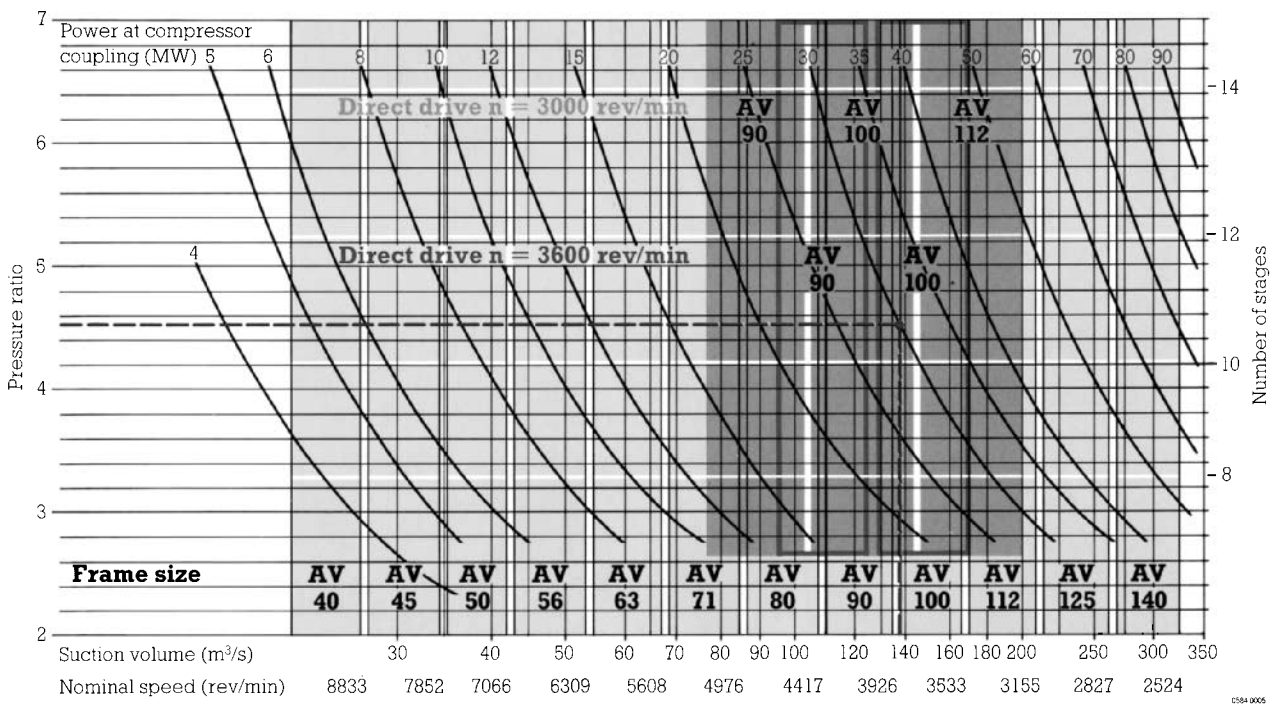
FIG. C-145 Determination of the discharge temperature t_2. (Source: Sulzer-Burckhardt.)

Figs. C-146 and C-147 are valid for air at suction conditions 1 bar, 20°C, 70% relative humidity.



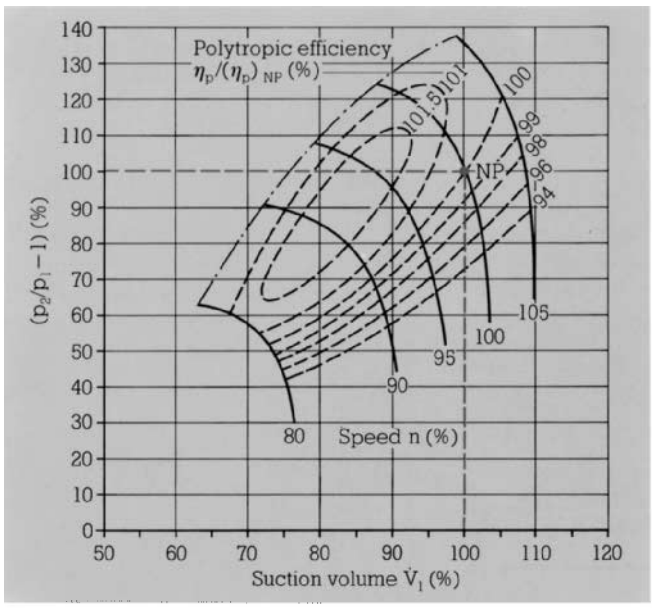
Selection of the compressor size: nominal diameter D (cm) = frame size as a function of the actual wet suction volume flow $\dot{V}_f$ (m³/s). Determination of the approximate number of stages z, nominal speed n (rev/min) and power input P (MW).

FIG. C-146 Performance data for type A (FIXAX). (Source: Sulzer-Burckhardt.)



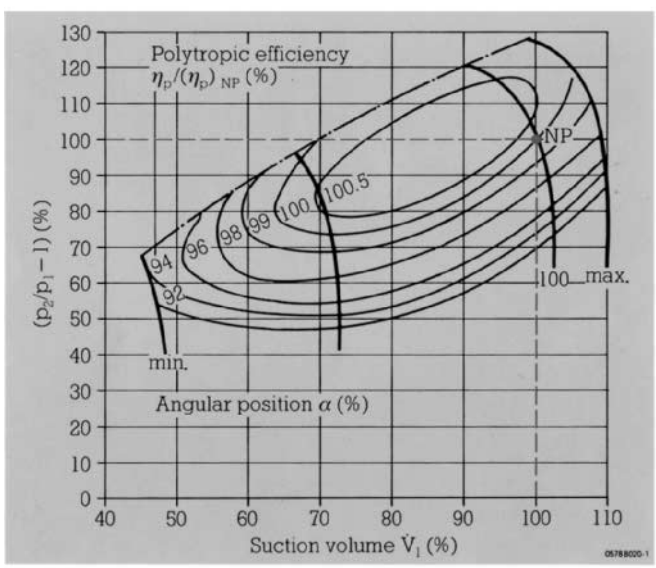
Selection of the compressor size: nominal diameter D (cm) = frame size as a function of the actual wet suction volume flow $\dot{V}_f$ (m³/s). Determination of the approximate number of stages z, nominal speed n (rev/min) and power input P (MW).

FIG. C-147 Performance data for type AV (VARAX). (Source: Sulzer-Burckhardt.)



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Operating characteristics for a compressor, type A, with fixed stator blades and variable speed:
NP = Reference point (100%) = Design point;
p_2/p_1 = Pressure ratio.
Valid for: constant gas data, constant inlet temperature.



149

Operating characteristics for a compressor, type AV, with adjustable stator blades and constant speed:
NP = Reference point (100%) = Design point;
p_2/p_1 = Pressure ratio;
α = Angular setting of the stator blades.
Valid for: constant gas data, constant inlet temperature.

Depending on the specific process requirements, such as extremely high turn-down, maximum efficiency within a certain range, flat or steep p-V curve, etc., various parameters of the blading may be changed in order to adapt the characteristics to such special conditions.

Selection and performance calculation of an axial compressor

	Calculation example Air compressor, type A
Given	
Capacity	$\dot{m}_t = 146.2$ kg/s
Suction pressure	$p_1 = 1$ bar abs
Suction temperature	$T_1 = 308$ K, $t_1 = 35$ °C
Relative humidity	$\varphi_1 = 60\%$
Discharge pressure	$p_2 = 4.5$ bar abs
Dry molecular mass	$M_t = 28.95$ kg/kmol
Isentropic exponent c_p/c_v	$k = 1.4$
Compressibility factor	$Z = 1$
Calculation instructions	
1. Determination of the absolute humidity x and wet molecular mass M_f (from T_1, p_1, φ_1) with Fig. C-144	$x = 0.021$ $M_f = 28.6$ kg/kmol
2. Calculation of the wet mass flow $\dot{m}_f = \dot{m}_t(1 + x)$	$\dot{m}_f = 146.2\,(1+0.021)$ $= 149.3$ kg/s
3. Determination of the actual suction volume $\dot{V}_1 = \frac{\dot{m}_f \cdot T_1}{F^* \cdot M_f \cdot p_1}$ [m³/s]	$\dot{V}_1 = 133.7$ m³/s
4. Determination of the discharge temperature t_2 with Fig. C-145	$t_2 = 225$ °C
5. Selection of the compressor size (nominal diameter D) and number of stages Z, determination of speed n and power input P with Fig. C-146	D = 90 cm Z = 12 n = 3600 rev/min P = 29 MW
6. Type designation	A 90–12
7. Selection of compressor size AV with Fig. C-147	
7.1 Steam turbine drive, i. e. no speed restriction (speed $n_s = n_n \cdot \sqrt{Z_{th}/Z_s}$) $= 3926 \cdot \sqrt{10.6/11.0}$ Type designation	D = 90 cm Z = 11 n = 3854 rev/min P = 29 MW AV 90–11
7.2 Electric motor direct drive 3600 rev/min Type designation	D = 100 cm Z = 11 n = 3600 rev/min P = 29 MW AV 100–11
7.3 Electric motor direct drive 3000 rev/min $Z_s = Z_{th} \cdot \left(\frac{n_n}{n_s}\right)^2 = 10.6 \cdot \left(\frac{3533}{3000}\right)^2$ $= 14.7$ Type designation	D = 100 cm Z = 15 n = 3000 rev/min P = 29 MW AV 100–15

n_n = Nominal speed
n_s = Selected speed for electric motor direct drive or corrected to arrive at a whole number of stages
Z_{th} = Theoretical number of stages on the basis of Figs. C-146 and C-147
Z_s = Selected number of stages to arrive at a whole number or to account for reduced speed in example 7.3. The reduced rotational speed imposed by the speed of the two-pole motor reduces the tip speed proportionately. This must be compensated by increasing the number of stages to produce the desired pressure ratio.

* Factor F, if mass $\dot{m}$ given in Nm³/h = 33505
SCFM (14.7 psia, 60 F) = 20844
SCFM (14.7 psia, 70 F) = 21245
kg/s = 12.027

Performance data [type A (FIG. C-148), type AV (FIG. C-149)] for selection and performance calculation of an axial compressor. (Source: Sulzer-Burckhardt.)

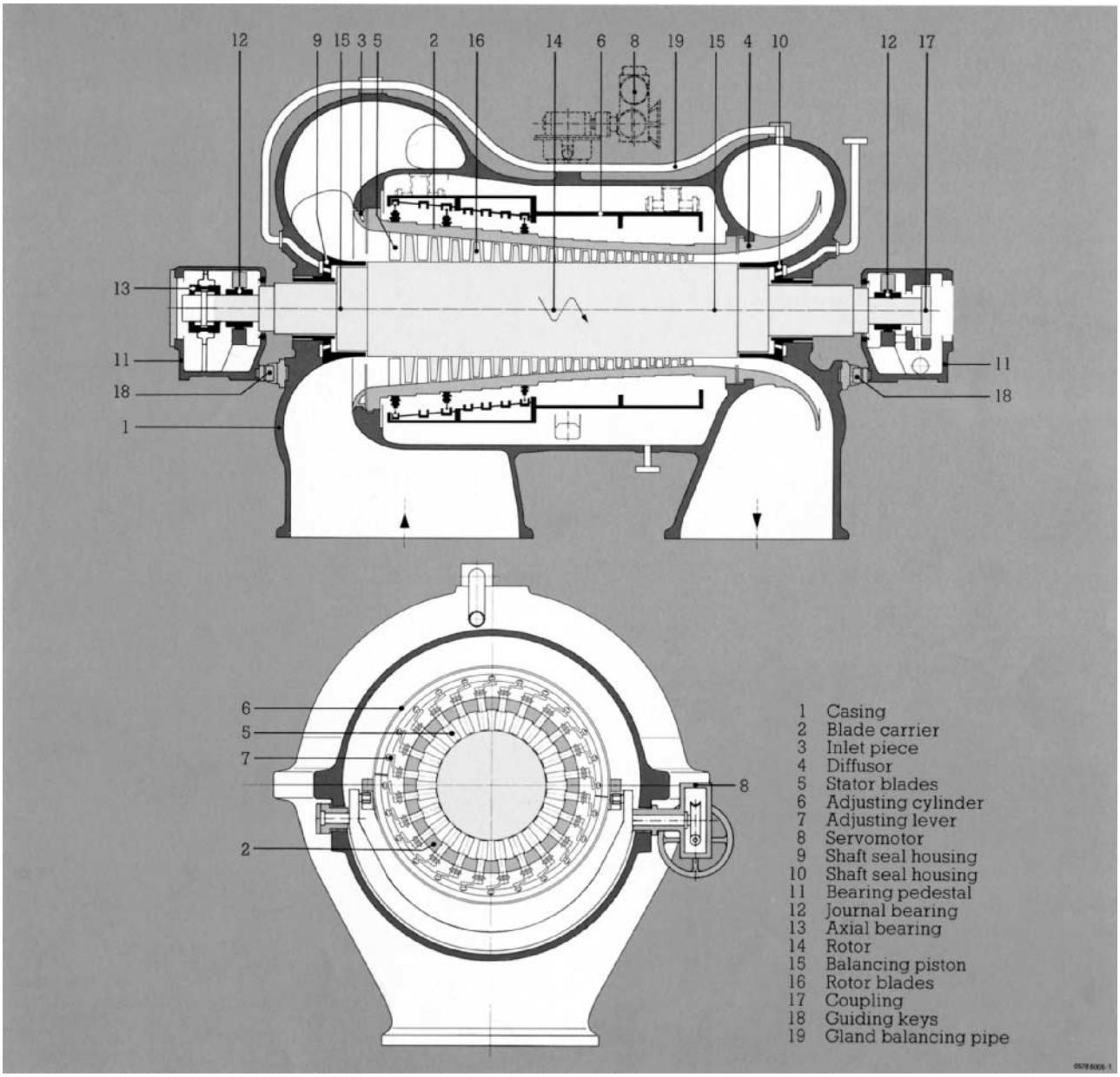


FIG. C-150 Longitudinal and cross-sectional view of an AV compressor. (Source: Sulzer-Burckhardt.)

can be specified. The suction branch may then be axial, or both suction and discharge may be routed upward or downward. In the vertical central plane, the casing is aligned by two keyways; it is equipped with four supporting feet. It is fixed at one end in the axial direction by one pair of feet. The other pair of feet of large frame sizes rests on pendulum supports with spherical contact surfaces. As a result of this, the casing can take up the thermal expansion in both the axial and lateral direction without difficulty. This feature is particularly advantageous in the case of

light steel foundations, as may be found, for instance, on an offshore platform. On small frame sizes the pendulum supports are replaced by sliding keyways.

Blade carrier, casing inserts. The blade carrier inserted in the casing is centered on both the suction and discharge side, and is able to expand freely in the axial direction. The diffusor and the gland inserts are also fitted as separate parts in the casing.

The double-casing design with outer casing and blade carrier offers various advantages:

- Rigid casing construction; the clearances in the blade duct are not influenced directly by external pipe forces.
- Simple fitting of the blades and assembly of the casing parts; the top half of the casing can be raised without dismantling the blade-adjusting mechanism.
- Possibility of fitting different blade carriers, for adapting the blade duct and thus the compressor characteristics to greatly changed operating conditions.
- Optimal protection of the adjusting mechanism in the space between the casing and blade carrier; the space is kept under suction pressure to safeguard the adjusting mechanism against condensation and corrosion attack.

Rotor. The rotors are usually of forged solid design. In the case of larger machines or if the moment of inertia must be minimized to limit the power requirement when running up with an electric motor, welded hollow rotors may be used. Integrated balancing pistons at both ends of the rotor facilitate an equalization of the axial thrust. The careful balancing of the rotor at full speed results in highly smooth running characteristics. If necessary, balancing can also be effected in the casing.

The labyrinth strips are caulked in the rotor.

Blading. Blading with a high degree of reaction, i.e., the increase in pressure takes place exclusively in the impellers, is employed for the compression of lighter gases such as helium or hydrogen.

For all other applications, such as the compression of air, blading with a lower percentage reaction is adopted. The increase in pressure is distributed to the rotor and stator blades. This enables the following design advantages to be realized:

- Higher efficiency with lower aerodynamic loading
- Widest possible control range with high part-load efficiency at constant speed
- Largest possible suction volume at given speed
- Increased reliability of operation due to larger radial blade clearances and the omission of guide vane sealings
- Steeper pressure volume characteristics, especially suitable for capacity control, for the parallel operation of different compressors, for refrigeration processes, and for exact adjustment of the blow-off line

The rotor and stator blades are normally made of 13 percent chrome steel and machined. When handling highly contaminated aggressive air or corrosive gases, alloys with higher chromium and nickel content may be used. The rotor blades have rhomboidal roots and are firmly braced in an exactly defined position in the peripheral grooves of the rotor. This is of particular importance for their vibration-wise design. The fixed stator blades are provided with a rectangular foot. The adjustable stator blades are made in one piece with a cylindrical shaft. The latter is seated in a bearing bush in the blade carrier. The high damping characteristics of this seating arrangement practically eliminate the potential for high vibration amplitudes associated with the stator blades.

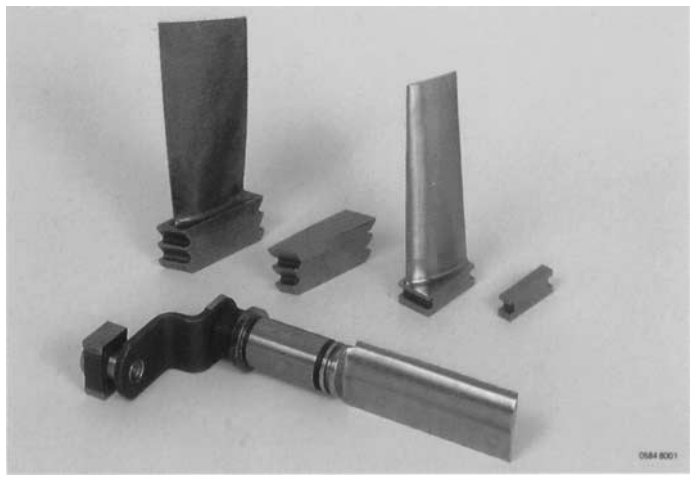


FIG. C-151 Adjustable stator blade, rotor blade, and fixed stator blade with intermediate piece. (Source: Sulzer-Burckhardt.)

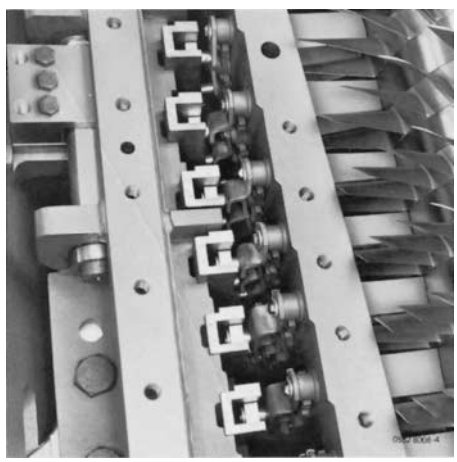


FIG. C-152 Stator blade adjusting mechanism. (Source: Sulzer-Burckhardt.)

Stator blade adjusting mechanism for the AV types. The adjusting mechanism is located in an annular space between casing and blade carrier, and is thus well protected against contaminants and moisture. It is maintenance-free and does not require any lubrication. (See Figs. C-151 and C-152.)

Servomotors. Two types of servomotors are available, electric and hydraulic (Figs. C-153 and C-154).

Automatic mass flow, volume or pressure control. The adjusting mechanism is operated by means of two hydraulic servomotors that are affixed laterally to the casing. One of the servomotors is equipped with a positioning transmitter and the second operates hydraulically in parallel.

The linear movement of the servomotor piston rods is transmitted directly to the adjusting cylinder by way of two ball and socket joints.

Remote setting of reference value. One single electric servomotor is attached laterally to the bottom half of the casing. Its driving shaft actuates a pivoted fork

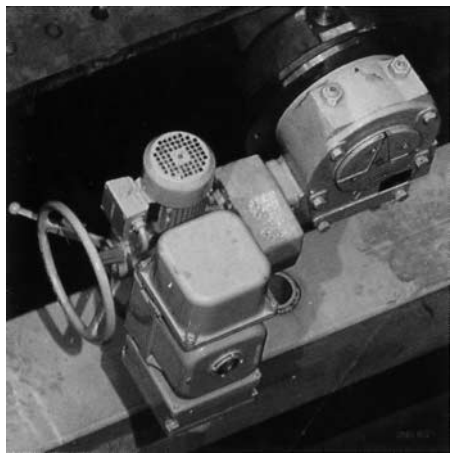


FIG. C-153 Electric servomotor. (Source: Sulzer-Burckhardt.)

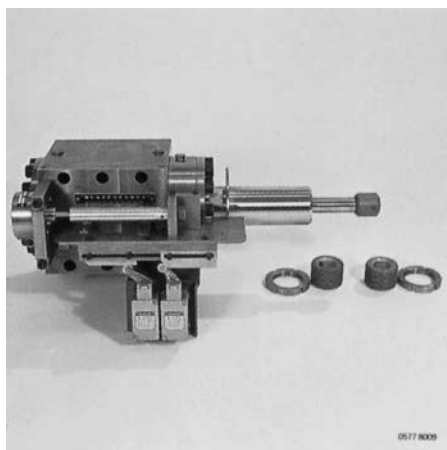


FIG. C-154 Hydraulic servomotor. (Source: Sulzer-Burckhardt.)

positioned on either side of the casing in maintenance-free bearings. This fork in turn moves the adjusting cylinder in the axial direction.

Adjusting cylinder. The adjusting cylinder of welded design can move in the axial direction and is dry-seated. There is no restriction of heat expansion in any direction. U-shaped guide rings are provided on the inner side in which the adjusting levers are engaged.

Stator blades. The adjusting levers provided on the end of each stator blade shaft are connected to the guide rings of the adjusting cylinder by means of pivoting sliders. The axial movement of the cylinder is converted into a rotating movement of the stator blades.

The self-lubricating bearing bushes of the blade shafts are seated in the radial holes of the blade carrier. O-ring packings prevent the ingress of contaminants into the stator blade seating.

Shaft seals. Labyrinth seals are used for the standard models. The stainless steel labyrinth strips are caulked on the rotor and are replaceable. In case of rubbing

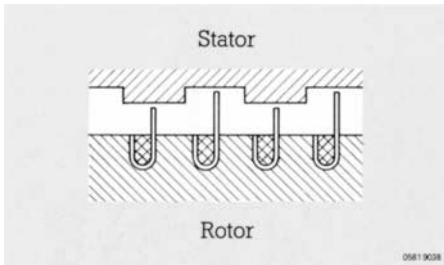


FIG. C-155 Labyrinth shaft seals. (Source: Sulzer-Burckhardt.)

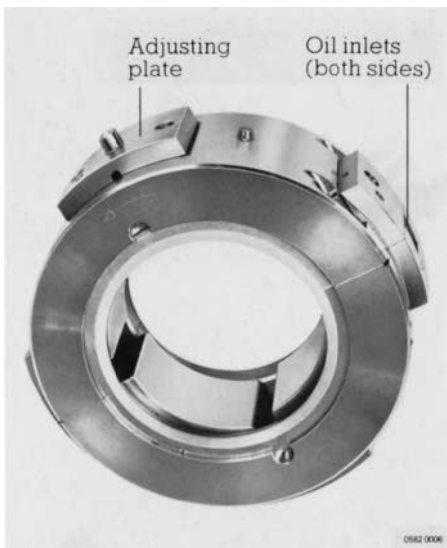


FIG. C-156 Multisegment journal bearing with four tilting pads. (Source: Sulzer-Burckhardt.)

due to unbalance, the friction-induced heat is immediately passed to the massive stator, thus avoiding distortion of the rotor. Gas-tight shaft seals and standstill seals can be fitted for special requirements. See Fig. C-155.

Journal and axial bearings

Journal bearings. In the standard version, i.e., with the compressor rotor solidly coupled and the rotor thrust transferred to the axial thrust bearing of the prime mover or the gear, the bearing housings are equipped only with journal bearings. Two-lobe bearings are provided for the lower speed range; tilting pad journal bearings are generally used for the higher speeds of the smaller frame sizes for reasons of stability. The slight curvature of the adjusting plates allows the bearings to be set accurately on erection. The bearings are firmly held in position by the bearing housing top half. See Figs. C-156 through C-158.

Two-lobe bearings are suitable for both senses of rotation, while tilting pad bearings are essentially for only one direction, although they can tolerate running backward with a somewhat reduced load capacity.

Axial thrust bearings. If a flexible coupling is selected between driver and driven machine, the bearing housing can accommodate the necessary tilting pad thrust bearing. The purpose of this bearing is to absorb the remaining thrust of the machine and any significant axial friction thrust of the coupling due to sharp

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